Linear Algebra

Linear Transformations

1 Linear transformations

A function $T\colon \mathbb{R}^n \to \mathbb{R}^m$ is a linear transformation if $\forall \vec{v}, \vec{u} \in \mathbb{R}^n$ and $\forall r \in \mathbb{R}$ and has the following properties:

(i) $T(\vec{u} + \vec{v}) = T(\vec{u}) + T(\vec{v})$

(ii) $T(r\vec{v}) = rT(\vec{v})$

Example: $f(x) = kx$ is a linear function $\mathbb{R}^1 \to \mathbb{R}^1$. The following properties hold:

$$f(x_1 + x_2) = k(x_1 + x_2) = f(x_1) + f(x_2)$$
$$f(rx_1) = k(rx_1) = rkx_1 = rf(x_1)$$

Second example: $f(x) = x^2$ is not a linear function $\mathbb{R}^1 \to \mathbb{R}^1$ a.k.a.

$$f(2x) \neq 2f(x)$$

LHS: $f(2x) = (2x)^2 = 4x^2 = 4f(x)$

RHS: $2f(x) \Rightarrow$ LHS $\neq$ RHS

Remark. To show that a function is linear, you can combine (i) and (ii) together.

$$(i) + (ii) = f(\vec{u} + r\vec{v}) = f(\vec{u}) + rf(\vec{v}) \ \forall \vec{u}, \vec{v} \in \mathbb{R}^n, \forall r \in \mathbb{R}$$

Capital letters may be used to denote a linear function. Don't confuse them and matrices!

1.1 What are functions?

A function *compacts* complex statements. For example:

Let $h(x) = 4x^2 + 6x$

Then we can write $4 + h(x)$ as $4 + 4x^2 + 6x$, and anytime we see $4x^2 + 6x$ appear in a statement, we can convert it back to $h(x)$. Note that $h(x)$ doesn't have to be represented in function notation; if we let $y = h(x)$, then we could write $4 + h(x)$ as $4 + y$. Knowing this is essential to prevent misunderstandings. That being said, if we define $g(x) = 1$ then whenever $g(x)$ appears anywhere in a statement we can replace it with 1 (isn't that just a constant?).

1.2 What are sets?

Sets are an unordered collection of anything. For example, $\{1, 2, 3, 4\}$ is a set. We can also have a set of real numbers, or a set of even numbers, which would be denoted as $\{\, x \in \mathbb{Z} \mid x \,\%\, 2 = 0 \,\}$. That's set builder notation, and here's how it works: {our set | if it meets what we write here}. Knowing this will help mitigate confusion further on.

1.3 Matrices as a form of a linear function

If these properties hold, then $f(\vec{x})$ is linear. A is a matrix. Remember that $A\vec{x}$ returns a vector.

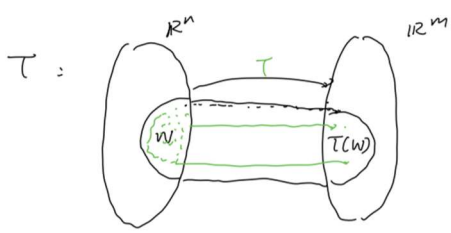
$$\begin{aligned}
&f(\vec{x}) = A\vec{x} = \vec{y} \Rightarrow f\colon \mathbb{R}^n \to \mathbb{R}^m \\
&A(\vec{x} + \vec{y}) = A\vec{x} + A\vec{y} \\
&A(r\vec{x}) = rA\vec{x}, r \in \mathbb{R} \\
&\Rightarrow f(\vec{x}) \text{ is linear}
\end{aligned}$$

1.4 Domain, Codomain, Image, and Kernel

A linear transformation's domain, codomain, image, and kernel are sets, and thus can be described in the set builder notation.

Definition: If $T\colon \mathbb{R}^n \to \mathbb{R}^m$ is a linear transformation then…

1. $\mathbb{R}^n$ is the domain of T. (The set of inputs)
2. $\mathbb{R}^m$ is the codomain of T. (The set of possible outputs)
3. Let $w \subseteq \mathbb{R}^n$, then the image of w under T is $T(w) = \{T(\vec{w}) | \vec{w} \in W\}$



   a. $T(w)$ is a collection of all possible outputs corresponding to the inputs of w. $T(w)$ is the image.

4. The range of T is $T(\mathbb{R}^n) = \{T(\vec{v}) | \vec{v} \in \mathbb{R}^n\}$ (i.e., Range of T is the image of the domain.)
5. Let $w' \subseteq \mathbb{R}^m$. The inverse image of w' under T is $T^{-1}(w') = \{\vec{v} \in \mathbb{R}^n | T(\vec{v}) \in w'\}$ (is this the domain?)
6. Let $\vec{0}$ be the zero vector in $\mathbb{R}^m$. $T^{-1}(\vec{0}) = \left\{\vec{v} \in \mathbb{R}^n \middle| T(\vec{v}) = \vec{0} \in \mathbb{R}^m\right\}$ is called the kernel of T. **(The *kernel* of a linear transformation T is the set of inputs that T sends to the zero vector.)**

1.5 An example of a linear transformation

Example: Let $T\colon \mathbb{R}^3 \to \mathbb{R}^2$. $T([x_1, x_2, x_3]) = [x_1 - x_2, x_2 + x_3]$.

This is one way for us to define a function and is a definition of a function.

Here's a series of questions:

1. **DETERMINING WHETHER T IS LINEAR**

   Is T a linear map/function/transformation? **Answer: Yes. Why?**
   $T(\vec{u} + r\vec{v}) = T(\vec{u}) + rT(\vec{v})\ \forall \vec{u}, \vec{v},\ r \in \mathbb{R}$.
   If we verify if LHS of the above equation is always equal to RHS, **then** T is a linear function. Our objective is to start with $T(\vec{u} + r\vec{v})$ and perform some operations to get us to $T(\vec{u}) + rT(\vec{v})$.
   Let $\vec{u} = [u_1, u_2, u_3], \vec{v} = [v_1, v_2, v_3] \in \mathbb{R}^3 = domain(T)$

   $$r \in \mathbb{R}$$

   Rationale: $\vec{u}$ and $\vec{v}$ are inputs to T and must be $\mathbb{R}^3$.

   LHS:
   $$\begin{aligned} T(\vec{u} + r\vec{v}) &= T([u_1, u_2, u_3] + r[v_1, v_2, v_3]) \\ &= T([u_1 + rv_1, u_2 + rv_2, u_3 + rv_3]) \end{aligned}$$

   Running the function, we get this. We can do some algebraic manipulation to get what we have as close to the RHS,
   $$\begin{aligned} &= [u_1 + rv_1 - u_3 - rv_3, u_2 + rv_2 + u_3 + rv_3] \\ &= [u_1 - u_3 + r(v_1 - v_3), u_2 + u_3 + r(v_2 + v_3)] \\ &= [u_1 - u_3, u_2 + u_3] + [r(v_1 - v_3), r(v_2 + v_3)] \\ &= [u_1 - u_3, u_2 + u_3] + r[v_1 - v_3, v_2 + v_3] \end{aligned}$$

   Recall:
   $$\begin{aligned} T(x_1, x_2, x_3) &= (x_1 - x_3, x_2 + x_3) \\ &= T([u_1, u_2, u_3]) + rT([v_1, v_2, v_3]) \end{aligned}$$

   By definition:

   $$T(\vec{u}) + rT(\vec{v}) = \text{RHS}$$

   Therefore LHS = RHS $\Rightarrow T$ is a linear transformation.

2. **FINDING THE IMAGE OF A FUNCTION WITH AN INPUT DOMAIN**

   Let $H = \{[x, x, x] \in \mathbb{R}^3\} \subseteq \mathbb{R}^3$. Find the **image** of H under T.

   Recall: We let $T\colon \mathbb{R}^3 \to \mathbb{R}^2$. $T([x_1, x_2, x_3]) = [x_1 - x_2, x_2 + x_3]$.

   Objective: Want to find the image of H under T in the most explicit way as possible.

   $$T(H) = \{\vec{y} \in \mathbb{R}^2 | \vec{y} = T(\vec{u}), \vec{u} \in H\}$$

   The above notation says $\vec{y}$ is in $\mathbb{R}^2$ (because $T(H)$ is in $\mathbb{R}^2$), but only if $\vec{y}$ is an output of $T(\vec{u})$ where $\vec{u}$ is in H, a set of $\mathbb{R}^3$ vectors where all its entries are identical. We *could* stop at this step, but why not expand this further?

   $$\begin{aligned} &= \{\vec{y} \in \mathbb{R}^2 | \vec{y} = T([u, u, u]), \vec{u} = [u, u, u] \in H\} \\ &= \{\vec{y} \in \mathbb{R}^2 | \vec{y} = [0, u + u] = [0, 2u], u \in \mathbb{R}\} \end{aligned}$$

   Conclude that $\{\vec{y} \in \mathbb{R}^2 | \vec{y} = [0, 2u], u \in \mathbb{R}\}$

This is the image of H under T. This is an explicit representation of an image without $T(\vec{u})$ appearing anywhere within the set builder.

3. **DEDUCING INVERSE IMAGES WITH AN IMAGE**

Question: Let $u = \{[1,2],[-1,3]\} \subseteq \mathbb{R}^2$ (which is a subset of the codomain of T. Find the inverse image of u, under T.

We are given a **set** u with two elements – it's like two dots in a set, so what we actually have to do is compute a function's inverse with these inputs – we were given two inputs, which means double the work!

Recall: We let $T\colon \mathbb{R}^3 \to \mathbb{R}^2$. $T([x_1,x_2,x_3]) = [x_1 - x_2, x_2 + x_3]$.

The set builder below denotes our inverse image. We just have to compact that down, so we end up with an actual set.

$$\begin{aligned} T^{-1}(u) &= \{\vec{x} = [x_1,x_2,x_3] \in \mathbb{R}^3 |\ T(\vec{x}) \in u\} \\ &\Rightarrow T(\vec{x}) = [1,2] \text{ or } [-1,3] \end{aligned}$$

$T(\vec{x})$ has two different equivalences. We need to consider both.

Case 1: $T(\vec{x}) = [1,2]$. You'll need to find $[x_1,x_2,x_3]$ such that $T([x_1,x_2,x_3]) = [1,2]$

$$\begin{aligned} &[x_1 - x_3, x - 2 + x_3] = [1,2] \Rightarrow \begin{matrix} x_1 - x_3 = 1 \\ x_2 + x_3 = 2 \end{matrix} \\ &\Rightarrow \left[\begin{array}{ccc|c} 1 & 0 & -1 & 1 \\ 0 & 1 & 1 & 2 \end{array}\right] \end{aligned}$$

So, you might be asking, where did the augmented matrix come from? That was because a system of linear equations appeared in your work, so you must solve it. It is a 3-unknown 2-equation system, so you will end up with a free variable.

3-variable, 2-equation systems typically have infinite solutions, and can be solved like any other linear system. If you're having trouble making sense out of this, you can always add a useless equation $0x_1 + 0x_2 + 0x_3 = 0$ giving you

$$\left[\begin{array}{ccc|c} 1 & 0 & -1 & 1 \\ 0 & 1 & 1 & 2 \\ 0 & 0 & 0 & 0 \end{array}\right]$$

Which will have a free variable, resulting in infinitely many solutions.

This will give you: $\begin{cases} x_1 - x_3 = 1 \\ x_2 + x_3 = 2 \end{cases} \Rightarrow \begin{matrix} x_3 = s \\ x_1 = 1 + s \\ x_2 = 2 - s \end{matrix}$

We can thus conclude that:

$$\vec{x} = [x_1,x_2,x_3] = [1+s, 2-s, s], s \in \mathbb{R}$$

We deduced a part of the inverse image of u under T.

$$T\left(\left[\frac{1+s}{x_1}, \frac{2-s}{x_2}, \frac{s}{x_3}\right]\right) = \left[\frac{1+s}{x_1} - \frac{s}{x_3}, \frac{2-s}{x_2} + \frac{s}{x_3}\right] = [1,2]$$

Case 2: $T(\vec{x}) = [x_1 - x_3, x_2 + x_3] = [-1,3]$

$$\Rightarrow \begin{cases} x_1 - x_3 = -1 \\ x_2 + x_3 = 3 \end{cases} \Rightarrow \vec{x} = [-1+t, 3-t, t], t \in \mathbb{R}$$

That's case 2, complete. Now we can combine case 1 and 2 together and that is our inverse image, which is:

$$[1+s, 2-s, s] \text{ and } [-1+t, 3-t, t]$$

Or we can represent it as a set: $\{[1+s, 2-s, s], [-1+t, 3-t, t]\}$

Because u is an "image", we can represent the inverse image like the following:

$$T(\{\, [1+s, 2-s, s], [-1+t, 3-t, t] \mid t, s \in \mathbb{R} \,\}) = u$$

$$T\left(\left\{ \begin{bmatrix} 1+s \\ 2-s \\ s \end{bmatrix}, \begin{bmatrix} -1+t \\ 3-t \\ t \end{bmatrix} \middle| t, s \in \mathbb{R} \right\}\right) = u$$

4. **FINDING THE KERNEL OF A FUNCTION**

   Find the kernel of T.

$$\ker(T) = \{\vec{x} \in \mathbb{R}^3 | T(\vec{x}) = 0\}$$

   Figure out when input $\vec{x}$ can give you 0 as its output.

   Recall: $T(\vec{x}) = T([x_1, x_2, x_3]) = [x_1 - x_3, x_2 + x_3] = [0, 0]$

   This can be used to build a system of equations.

$$\left[\begin{array}{ccc|c} 1 & 0 & -1 & 0 \\ 0 & 1 & 1 & 0 \end{array}\right] \Rightarrow [s, -s, s], s \in \mathbb{R}$$
$$T([s, -s, s]) = [s - s, -s + s] = [0, 0] \Rightarrow \vec{0} \in \mathbb{R}^2$$
$$\ker(T) = \{[s, -s, s] \in \mathbb{R}^3 | s \in \mathbb{R}\}$$

1.6 Basis vectors

Proposition: Let $T\colon \mathbb{R}^n \to \mathbb{R}^m$ be a linear transformation. Then

1. If
   $\vec{v}_1, \vec{v}_2, \dots, \vec{v}_k \in \mathbb{R}^n \Rightarrow r_1, r_2, \dots, r_k \in \mathbb{R}$
   $$\begin{aligned} &T(r_1\vec{v}_1 + r_2\vec{v}_2 + \cdots + r_k\vec{v}_k) \\ &= rT(\vec{v}_1) + r_2T(\vec{v}_2) + \cdots + r_kT(\vec{v}_k) \end{aligned}$$
   This is a linear combination between $\overrightarrow{v_1}, \cdots, \vec{v}_k$

Remark: What if $\overrightarrow{v_1} \cdots \overrightarrow{v_n}$ is a set of **basis** vectors of $\mathbb{R}^n$, i.e., $span(\overrightarrow{v_1} \cdots \overrightarrow{v_n}) = \mathbb{R}^n$ and $\vec{v}_1 \cdots \vec{v}_n$ are linearly independent

Then

$$\Leftrightarrow \begin{bmatrix} | & | & | \\ \vec{v}_1 & \cdots & \vec{v}_n \\ | & | & | \end{bmatrix} \sim \begin{bmatrix} 1 & \cdots & 0 \\ \vdots & \ddots & \vdots \\ 0 & \cdots & 1 \end{bmatrix}$$

Proposition: For any input $\vec{\alpha} \in \mathbb{R}^n. \vec{\alpha} = r_1\overrightarrow{v_1} + r\overrightarrow{v_2} + \cdots + r\overrightarrow{v_n}$ (given $\overrightarrow{v_1} \cdots \overrightarrow{v_n}$ are a set of basis vectors)

$$T(\vec{\alpha}) = r_1 T(\overrightarrow{v_1}) + r_2 T(\overrightarrow{v_2}) + \cdots + r_n T(\vec{v}_n)$$

Tell me the image of the basis vectors without telling me the formula of T, and I would still be able to compute the value of the output for any given no input. If you have the image of basic for the T, then you can compute the output for any given input.

Property: The linear transformation must fix a zero. This means $T(\vec{0}) = \vec{0}$ must be in the codomain. This gives us a shortcut to check if a given function is linear or not. If a zero gives an output as a nonzero vector, then you can tell it is not a linear transformation.

Proof. Let $\vec{0} = \vec{v} - \vec{v}. \vec{v} \in \mathbb{R}^n$.

$$T(\vec{0}) = T(\vec{v} - \vec{v}) = T(\vec{v}) - T(\vec{v}) = \vec{0}$$

$\Rightarrow$ ker (T) always contains the zero vector.

2 Vector Spaces and Linear Independence

A homogeneous linear system can be represented as:

$$A\vec{x} = 0$$

Where A is an $m \times n$ matrix, where $\vec{x} = \begin{bmatrix} x_1 \\ \vdots \\ x_n \end{bmatrix}$.

$$T: \mathbb{R}^n \to \mathbb{R}^m: T(\vec{x}) = A\vec{x}$$

For homogeneous system $T(\vec{x}) = 0$, $\vec{x}$ is the kernel of T.

Remark

1. Every homogeneous system is consistent (≥ 1 soln)

   If the system gives you no solutions – you know how that looks like. But, for a homogeneous equation, the right side of a partitioned matrix will always be zero for all components.

   Moreover, the trivial solution always is a solution for $A\vec{x} = \vec{0}$. The reason why we call this the trivial solution is that $A(\vec{0}) = \vec{0}, \vec{x} = 0$

2. If $A \sim H$ and H has a pivot in every column, then $A\vec{x} = 0$ only has the trivial solution ($\vec{x} = \vec{0}$).

3. If $m < n$ (number of rows $<$ columns) and $A \sim H$ (H is in RREF) then it is impossible for H to have a pivot in every column, which means a free variable definitely exist. This means $A\vec{x} = \vec{0}$ has infinitely many solutions.

2.1 Null Space, row space, col space

Definition. Let A be an $m \times n$ matrix

1. The null space of A is a set of vectors $\vec{x}$ such that $A\vec{x} = 0$ i.e., $\{\, \vec{x} \in \mathbb{R}^n \mid A\vec{x} = 0 \,\}$. It is the set of vectors that go to the zero vector when multiplying A from the right.
2. The row space of A, $row(A)$ is the span of row vectors of A.
3. The column space of A, $col(A)$ is the span of column vectors of A.

A linear system $A\vec{x} = \vec{b}$ has a solution if and only if $\vec{b}$ is in the column space of A.

2.1.1 Deducing Null Spaces

For example, suppose I have this matrix:

$$A = \begin{bmatrix} 2 & -6 & -2 & 4 \\ -1 & 3 & 3 & 2 \\ -1 & 3 & 7 & 10 \end{bmatrix}$$

$$\begin{aligned} row(A) &= span([2, -6, 2, 4], [-1, 3, 3, 2], [-1, 3, 7, 10]) \\ col(A) &= span\left(\begin{bmatrix} 2 \\ -1 \\ -1 \end{bmatrix}, \begin{bmatrix} -6 \\ 3 \\ 3 \end{bmatrix}, \begin{bmatrix} -2 \\ 3 \\ 7 \end{bmatrix}, \begin{bmatrix} 4 \\ 2 \\ 10 \end{bmatrix}\right) \\ null(A) &= \{\, \vec{x} \in \mathbb{R}^4 \mid A\vec{x} = 0 \,\} \end{aligned}$$

We need to solve $A\vec{x} = 0$.

$$\left[\begin{array}{cccc|c} 2 & -6 & -2 & 4 & 0 \\ -1 & 3 & 3 & 2 & 0 \\ -1 & 3 & 7 & 10 & 0 \end{array}\right] \sim \left[\begin{array}{cccc|c} 1 & -3 & 0 & 4 & 0 \\ 0 & 0 & 1 & 2 & 0 \\ 0 & 0 & 0 & 0 & 0 \end{array}\right]$$

There are infinitely many solutions to this.

$$\begin{aligned} &x_2 = t \\ &x_4 = s \\ &x_3 + 2x_4 = 0 \Rightarrow x_3 = -2x_4 = -2s \\ &x_1 - 3x_2 + 4x_4 = 0 \Rightarrow x_1 = 3x_2 - 4x_4 = 3t - 4s \end{aligned}$$

$$\begin{aligned} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \\ x_4 \end{bmatrix} &= \begin{bmatrix} 3t - 4s \\ t \\ -2s \\ s \end{bmatrix} = \begin{bmatrix} 3t \\ t \\ 0 \\ 0 \end{bmatrix} + \begin{bmatrix} -4s \\ 0 \\ -2s \\ s \end{bmatrix} \\ &= t \begin{bmatrix} 3 \\ 1 \\ 0 \\ 0 \end{bmatrix} + s \begin{bmatrix} -4 \\ 0 \\ -2 \\ 1 \end{bmatrix} \end{aligned}$$

Therefore:

$$null(A) = span\left(\begin{bmatrix}3\\1\\0\\0\end{bmatrix}, \begin{bmatrix}-4\\0\\-2\\1\end{bmatrix}\right)$$

Null space is the only space that requires computations.

2.1.2 Proofs regarding null spaces

Proposition: Let $A\vec{x} = \vec{b}$ be a linear system with a particular solution p. ($A\vec{p} = \vec{b}$)

If $\vec{h}$ is a vector in the $null(A)$, then $\vec{p} + \vec{h}$ is also a solution for $A\vec{x} = \vec{b}$.

Proof. $A(\vec{p} + \vec{h}) = \vec{b}$

$$\begin{aligned}LHS &= A\vec{p} + A\vec{h}\\ LHS &= A\vec{p} + 0\\ LHS &= \vec{b}\end{aligned}$$

Remember that because h is a vector in the null space of A, it means $A\vec{h}$ is always equals to zero. $A\vec{p} = b$ is true because it is defined as true as part of the proof.

If $\vec{q}$ is any solution to $A\vec{x} = \vec{b}$, then $\vec{q} = \vec{p} + \vec{h}_1$ for some $\vec{h} \in null(A)$

$$A\vec{q} = A(\vec{p} + \vec{h}_1) = A\vec{p} + A\vec{h}_1 = \vec{b} + 0 \Rightarrow \vec{h}_1 \in null(A)$$

2.2 Linear independence

Definition: Linear independence (every single vector is useful; none of the other vectors can be made up through linear combinations of other vectors).

If $\vec{v}_1, \vec{v}_2, \dots, \vec{v}_k \in \mathbb{R}^n$, $r_1\vec{v}_1 + r_2\vec{v}_2 + \cdots + r_k\vec{v}_k = \mathbf{0}$ has exactly 1 solution (that is $\vec{r}$ being all zero), then we say that $\vec{v}_1, \dots, \vec{v}_k$ are linearly independent.

(It is impossible that this system has no solutions. If this system has infinite solutions, then $\vec{v}_1, \vec{v}_2, \dots, \vec{v}_k$ are linearly dependent.)

2.2.1 Linear independence properties

Remark: $r_1 = r_2 = r_3 = \cdots = r_k = 0$ is a solution for $r_1\vec{v}_1 + r_2\vec{v}_2 + \cdots + r_k\vec{v}_k = 0$, and if that is the only solution then $\vec{v}_1, \vec{v}_2, \dots, \vec{v}_k$ are linearly independent.

Suppose not. $\alpha_1, \alpha_2, \dots, \alpha_k$ is another solution for $r_1\vec{v}_1 + r_2\vec{v}_2 + \cdots + r_k\vec{v}_k = 0$. $\exists \alpha_i \neq 0$ (Why are we writing that? Because one solution has $r_1 \dots r_k$ be all zero and we are trying to incorporate a non-all-zero solution for $r_1\vec{v}_1 + r_2\vec{v}_2 + \cdots + r_k\vec{v}_k = 0$. Our work is here:

$$\begin{aligned}&\alpha_1\vec{v}_1 + \alpha_2\vec{v}_2 + \cdots + \alpha_i\vec{v}_i + \cdots + \alpha_k\vec{v}_k = 0\\ &\alpha_1\vec{v}_1 + \alpha_2\vec{v}_2 + \cdots + \alpha_{i-1}\vec{v}_{i-1} + \cdots + \alpha_k\vec{v}_k = -\alpha_i\vec{v}_i\\ &\frac{\alpha_1}{\alpha_i}\vec{v}_1 + \frac{\alpha_2}{\alpha_i}\vec{v}_2 + \cdots + \frac{\alpha_k}{\alpha_i}\vec{v}_k = -\vec{v}_i\end{aligned}$$

If $\vec{v}_i$ can be represented by the left equation, then $\vec{v}_i$ is redundant, which is *impossible* because we said that $\vec{v}_1, \dots, \vec{v}_k$ are linearly independent. We thus arrive at a contradiction.

Definition: Two vectors in $\mathbb{R}^n$ are linearly independent if they are non-zero and nonparallel.

2.2.2 Testing for linear independence

Example: $\vec{v}_1 = [2, -1, -1], \vec{v}_2 = [-2, 3, 7], \vec{v}_3 = [2, 1, 5]$

Solution: If $r_1\vec{v}_1 + r_2\vec{v}_2 + r_3\vec{v}_3 = 0$ **has exactly one solution,** the vectors are linearly independent. You can represent this in a matrix $A\vec{r}$, where A's rows are formed by the column vectors $\vec{v}$.

If a column is missing a pivot, then it is impossible for this system to have a unique solution $\vec{r}$.

$$\begin{bmatrix} 2 & -2 & 2 \\ -1 & 3 & 1 \\ -1 & 7 & 5 \end{bmatrix} \sim \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 2 & 1 & 0 \end{bmatrix}$$

While we did not reduce the above matrix to RREF, but this is good enough as we can see there is at least one column without the pivot.

2.2.3 Testing for linear dependence

Because $r_1\vec{v}_1 + r_2\vec{v}_2 + r_3\vec{v}_3 = 0$ has more than one solution, the vectors are linearly dependent.

Is this matrix invertible (if no, then we've got linear dependence)?

$$\begin{bmatrix} 2 & -1 & -1 \\ -2 & 3 & 7 \\ 2 & 1 & 5 \end{bmatrix} \sim \begin{bmatrix} 1 & 0 & 2 \\ 0 & 1 & 1 \\ 0 & 0 & 0 \end{bmatrix}$$

The rightmost column causes this matrix to have a free variable. If we let it equal to zero:

$$\left[\begin{array}{ccc|c} 1 & 0 & 2 & 0 \\ 0 & 1 & 1 & 0 \\ 0 & 0 & 0 & 0 \end{array}\right]$$

$A\vec{x} = 0$ has the solution $r \begin{bmatrix} 2 \\ 1 \\ -1 \end{bmatrix}$ (we factored out the negatives)

This means $2[\vec{v}_1] + 1[\vec{v}_2] - 1[\vec{v}_3] = 0$

$$[\vec{v}_3] = 2[\vec{v}_1] + [\vec{v}_2]$$

$\vec{v}_3$ is a redundant vector, and so could the others?

3 Subspaces and basis vectors

Hint: checking "in a subspace" WILL appear on an exam.

Definition: Let $W \subseteq \mathbb{R}^n$. W is called **a subspace** of $\mathbb{R}^n$ if these following properties hold:

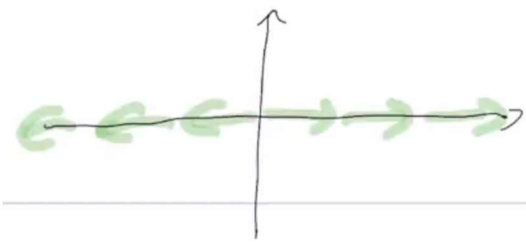
1. $W \neq \emptyset$
2. If $\vec{u}, \vec{v} \in W$, then $\vec{u} + \vec{v} \in W$ (Add two vectors in the "subspace" and your sum must still be within the subspace. This is the closure property regarding subspaces.)
3. If $\vec{u} \in W, r \in \mathbb{R}$, then $r \cdot \vec{u} \in W$ (Closure under scalar multiplication)

To check properties 2 and 3, you may need to pick variables arbitrarily and you must end up with a way to represent the sum or scalar product of the vectors in the way an individual vector in the set would be defined.

With all three holding W is a subspace.

3.1 Checking if a set is a subspace: Examples

Example: $W = \{\, \vec{v} \in \mathbb{R}^2 \mid \vec{v} = [x, 0], x \in \mathbb{R} \,\}$



Is W a subspace?

1. $W \neq \emptyset; \emptyset \subset W$ (NOT the zero vector.)
2. Pick $\vec{u} = [x_1, 0] \in W$, $\vec{v} = [x_2, 0] \in W$, The sum? $\vec{u} + \vec{v} = [x_1 + x_2, 0] \in W$ (it still lies on W and the x-axis.)
3. $r\vec{u} = r[x, 0] = [rx, 0] \in W$

Because these properties hold, W is a subspace in $\mathbb{R}^2$.

Second example: $H = \{\vec{v} \in \mathbb{R}^2 \mid \vec{v} = [0, y], y \in \mathbb{R}\}$

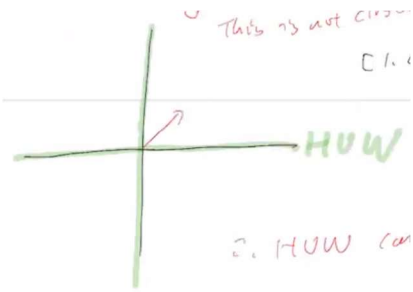
You can show that H is a subspace in $\mathbb{R}^2$.

Third example: $H \cup W$? No. The reason why this is not a subspace.

We can disprove this with vector addition:

$$[1, 0] + [0, 1] = [1, 1] \notin H \cup W$$

Therefore $H \cup W$ can't be a subspace.



Fourth example: (Try this out of lecture) $H \cap W$. That is just the zero vector. Scalar multiply a zero vector by any amount, and you will get the zero vector. Add the zero vector by another zero vector, and you get the zero vector. Therefore, the zero subspace is closed under addition and scalar multiplication.

Remark: I will check if $\vec{0}$ is contained in W **as a subspace must contain zero (but does not imply that a subspace is valid).**

Suppose W is a subspace. This implies that W is non-empty, implying $\exists \vec{u} \in W$. This also implies:

$$\begin{aligned} &\Rightarrow -1\vec{u} \in W \Rightarrow \vec{u} + (-1\vec{u}) \in W \\ &\Rightarrow \vec{0} \in W \end{aligned}$$

If $\vec{0}$ is not in the subspace, then I can immediately declare that W is not a subspace. Because all subspaces must be closed under addition and scalar multiplication you must be able to find zero when doing a combination of the two.

Example: $W = \text{sp}\{\vec{w}_1, \vec{w}_2, \dots, \vec{w}_k\}$, where $w_1, w_2, \dots, w_k \in \mathbb{R}^n \Rightarrow W$ is a subspace of $\mathbb{R}^n$.

Proof. Check if $\vec{0}$ belongs to W.

$$r_1\vec{w}_1 + r_2\vec{w}_2 + \cdots + r_k\vec{w}_k = \vec{0}$$

To show this is true, show that this system is homogeneous, and that there is always one solution: $r_1 = r_2 = \cdots = r_k = 0$.

Firstly, $\vec{0}$ is a vector in W, and a span means all possible linear combinations. Therefore, $\vec{0} \in W, \Rightarrow \vec{w} \neq 0$.

Step 2: Check if $\vec{u} + r\vec{v} \in W$ where $\vec{u} \cdot \vec{v} \in W, r \in \mathbb{R}$. If I can answer this, then $\vec{u}, \vec{v}$ are closed under vector addition and scalar multiplication.

$$\begin{aligned} \vec{u} &= \alpha_1\vec{w}_1 + \alpha_2\vec{w}_2 + \cdots + \alpha_k\vec{w}_k \\ \vec{v} &= \beta_1\vec{w}_1 + \beta_2\vec{w}_2 + \cdots + \beta_k\vec{w}_k \end{aligned}$$

Where $\alpha_1 \in \mathbb{R}\forall i, \beta_i \in \mathbb{R}\forall i$

$$\begin{aligned} \vec{u} + r\vec{v} &= (\alpha\vec{w}_1 + \cdots + \alpha_k\vec{w}_k) + (r\beta_1\vec{w}_1 + r\beta_2\vec{w}_2 + \cdots + r\beta_k\vec{w}_k) \\ &= (\alpha_1 + r\beta_1)\vec{w}_1 + (\alpha_2 + r\beta_2)\vec{w}_2 + \cdots + (\alpha_n + r\beta_n)\vec{w}_k \end{aligned}$$

This is just another linear combination, showing that $\vec{u} + r\vec{v} \in \text{sp}\{\vec{w}_1 \dots \vec{w}_k\} = W$ which means $\vec{u} + r\vec{v} \in W \Rightarrow W$ is closed under addition and scalar multiplication. Therefore, W is a subspace of $\mathbb{R}^n$.

3.2 Finding a basis for a set of vectors

In a nutshell:

If we need to find a basis for $W = \text{sp}(w_1, w_2, \dots, w_k)$

1. Form the matrix A whose jth column vector is w_j
2. Row-reduce A to REF, which we will call H.
3. The set of all w_j such that the jth column of H contains a pivot is a basis for W.

Definition: Let W be a subspace of $\mathbb{R}^n$.

If B $= \left\{\vec{b}_1, \vec{b}_2, \dots, \vec{b}_k\right\}$ is a subset of W, then B is a **basis** for W if every vector in W can be written **uniquely** as a linear combination of vectors in B. (All the vectors in W can be generated by B. If that holds, B has that property.)

Remark: Pick $\vec{w} \in W$.

$$\vec{w} = r_1\vec{b}_1 + r_2\vec{b}_2 + \dots + r_k\vec{b}_k, r_i \in \mathbb{R} \forall i$$

And we need to prove this representation is unique, i.e., $\{r_1, \dots, r_k\}$ is unique. This gives us another linear system, with unknowns $\vec{r}$. We need to check if this system has a unique solution; if it does, then those vectors $\vec{b}_1 \dots \vec{b}_k$ are basis vectors.

$$[\vec{b}_1 \quad \vec{b}_2 \quad \cdots \quad \vec{b}_k \mid \vec{0}]$$

This system having more than one solution indicates that $\{r_1 \dots r_k\}$ is not unique.

Example: $W = sp\{\vec{w}_1, \vec{w}_2, \dots, \vec{w}_5\}$

$$w_1 = \begin{bmatrix} 1 \\ 0 \\ 1 \\ 0 \end{bmatrix}, w_2 = \begin{bmatrix} 4 \\ 1 \\ 4 \\ 1 \end{bmatrix}, w_3 = \begin{bmatrix} 3 \\ 1 \\ 3 \\ 2 \end{bmatrix}, w_4 = \begin{bmatrix} -1 \\ 0 \\ -1 \\ 1 \end{bmatrix}, w_5 = \begin{bmatrix} -6 \\ 3 \\ -2 \\ 0 \end{bmatrix}$$

$$\begin{bmatrix} 1 & 0 & 0 & 0 & -6 \\ 0 & 1 & 0 & -1 & 3 \\ 0 & 0 & 1 & 1 & -2 \\ 0 & 0 & 0 & 0 & 0 \end{bmatrix}$$

Then only $\vec{w}_1, \vec{w}_2, \vec{w}_3$ are single-pivot row vectors; all of the other vectors are redundant. $\vec{w}_4 = -\vec{w}_2 + \vec{w}_3$ and $\vec{w}_5 = -6\vec{w}_1 + 3\vec{h}_2 - 2\vec{w}_3$

The basis vectors are $\text{sp}\{\vec{w}_1, \vec{w}_2, \vec{w}_3\}$.

This is equivalent to $\text{sp}\{\vec{w}_1, \vec{w}_2, \vec{w}_3, \vec{w}_4, \vec{w}_5\}$. The span is in $\mathbb{R}^3$; dimension$(\text{sp}\{\vec{w}_1, \vec{w}_2, \vec{w}_3\}) = 3$.

3.3 Basis Theorems

Theorem. Let $A \in M_{n\times n}$. The following are equivalent (if you can show one of them all in this list will be true):

1. The linear system $A\vec{x} = \vec{b}$ has a unique solution for any $\vec{b} \in \mathbb{R}^n$.
2. $A{\sim}I$
3. A is invertible.
4. The column vectors of A form a basis for $\mathbb{R}^n$.
5. The row vectors in A form a basis for $\mathbb{R}^n$.

3.3.1 Proofs for these Theorems

Proof. $1 \Rightarrow 2$ is the process of solving a linear system. Reducing A to RREF, if every single column has a pivot (n pivots), you'll get a unique solution. That matrix will look like I, thus forcing A to be row equivalent to the identity matrix.

This is trivially equivalent to 3. $[\,A \mid I\,]{\sim}[\,H \mid A^{-1}\,]$, where H is the identity matrix.

For no. 4: Column vectors form a basis $\Rightarrow$ they are linearly independent $\Rightarrow$ every column must contain a pivot $\Rightarrow$ you will have n pivots, meaning no free variables and a unique solution. If that is case, then $A{\sim}I \Leftrightarrow$ every single column has a pivot $\Leftrightarrow$ they can form a basis.

For no. 5: $A{\sim}I \Leftrightarrow A^T{\sim}I$

3.4 Rectangular cases

Theorem. Let A be $m \times n$ matrix. ($m > n$). The following are equivalent:

1. Each **consistent** system $A\vec{x} = \vec{b}$ has a unique solution.
2. The RREF of A must contain the $n \times n$ identity matrix with $m - n$ rows of zeros (on the bottom). This means there are $m - n$ redundant equations.
3. The column vectors of A form a basis for the column space of A.

3.4.1 Proofs for these theorems

Proof. $1 \Rightarrow 2$

$A\vec{x} = \vec{b}$, and for a unique solution, free variables are not allowed to exist, which means every column must contain a pivot. All entries below the identity matrix must be all zero, implying $m - n$ rows of zeroes.

Proof. $2 \Rightarrow 3$

Every column has a pivot $\Rightarrow$ the column vectors are linearly independent $\Rightarrow$ can serve as basis.

Proof. $3 \Rightarrow 1$

Basis $\Rightarrow$ linearly independent. Building a linear system out of the basis guarantees no free variables $\Rightarrow$ unique solution.

Remark. The number of solutions will depend on whether free variables exist. If you have a rectangular matrix $M_{m<n}$ (wider than it is tall), then $A\vec{x} = \vec{b}$ is a linear system with fewer equations than unknowns. If $A\vec{x} = \vec{b}$ is consistent (ruling out no solution situations), then it has infinitely many solutions.

Theorem. Let $B = \left\{\vec{b}_1, \dots, \vec{b}_k\right\}$ be a basis for the subspace $W \subset \mathbb{R}^n$, if and only if [1] $B = \left\{\vec{b}_1, \dots, \vec{b}_k\right\}$ is a linearly independent set and [2] it spans W.

Proof. B is a basis for $W \Leftrightarrow$ Pick any $\vec{w} \in W$; $\vec{w}$ has a unique representation: $\vec{w} = r_1\vec{b}_1 + r_2\vec{b}_2 + \cdots + r_k\vec{b}_k$, and the solution for r must be unique. You could view this as a linear system $A\vec{x} = \vec{b}$, that has a unique solution where $\vec{b} = \vec{w}$, and A is a matrix with column vectors $b_1 \dots b_k$, and $\vec{x} = [r_1, r_2, \dots, r_k] \Leftrightarrow$ no free variable $\Leftrightarrow$ every column must contain a pivot $\Leftrightarrow$ none of any element in B are redundant $\Leftrightarrow \forall \vec{b}_i$ (column vectors), it cannot be represented by linear combination of other $\vec{b}_i \Leftrightarrow$ they are linearly independent.

3.5 How to find a basis for any set of vectors

This: $W = \text{sp } \{\vec{w}_1 \cdots \vec{w}_k\}$ where $\vec{w}_i \in \mathbb{R}^n$ for $i = 1, 2, \dots, k$

Step 1. Create a matrix A where the columns of it are formed by the ith vectors in W.

Step 2: A~H, where H is REF or RREF. (Idea: check if every column has a pivot.

Step 3: The basis for W consists only of all the columns of A, corresponding to the columns of H that contains the pivots.

Definition: Let W be a subspace of $\mathbb{R}^n$. The number of elements in a basis for W is called the dimension of W, denoted by $\dim(w)$. There are two facts supported by this definition:

[EXAM HINT] Fact 1: Let W be a subspace of $\mathbb{R}^n$. Let $\vec{w}_1, \vec{w}_2, \dots, \vec{w}_k$ be vectors in W that span W and let $\vec{v}_1, \dots, \vec{v}_m$ be linearly independent vectors in W. Then, $k \geq m$. (This fact is trivial – the dimension of $W \geq K$.)

$\vec{v}_1 \cdots \vec{v}_m$ being linearly independent could be a basis for either W or a subspace of W. In other words, $\text{sp}\{\vec{v}_1 \cdots \vec{v}_m\} = U \leq W \Rightarrow m \leq k$.

Fact 2: Any two bases for a subspace $W \in \mathbb{R}^n$ contain the same number of vectors.

(The number of elements in a basis tells you the dimensions of the subspace it is in. If you have two bases with different numbers of vectors the dimensions formed by the span of it will be different.) For example, for $\mathbb{R}^2$: $\text{sp}([1,0],[0,1]) = \mathbb{R}^2$. Also, $\text{sp}([2,0],[0,2]) = \mathbb{R}^2$, as $\begin{bmatrix} 2 & 0 \\ 0 & 2 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} = \vec{b}$ has a unique solution, meaning every $\vec{b}$ can uniquely be represented by $[2,0],[0,2]$.
$\text{sp}([2,0],[0,2],[a,b])$, $[a,b] = \frac{a}{2}[2,0] + \frac{b}{2}[0,2] \Rightarrow [a,b]$ is a redundant vector.

Proving Fact 1: Given:

1. $\text{sp}\{\vec{w}_1 \cdots \vec{w}_k\} = W \in \mathbb{R}^n$.
2. $\vec{v}_1 \cdots \vec{v}_m \in W$ are linearly independent

Show $k \geq m$.

Suppose $k < m$. By the idea of span, $\forall \vec{v}_i \ (i \in [1, 2, 3, \ldots, m]) \in W = \text{sp}\{\vec{w}_1 \cdots \vec{w}_k\}$. All of v_i are chosen from W. This means

$$\begin{aligned}
\vec{v}_1 &= a_{11}\vec{w}_1 + a_{21}\vec{w}_2 + \cdots + a_{k1}\vec{w}_k \\
\vec{v}_2 &= a_{12}\vec{w}_1 + a_{22}\vec{w}_2 + \cdots + a_{k2}\vec{w}_k \\
&\vdots \\
\vec{v}_m &= a_{1m}\vec{w}_1 + a_{2m}\vec{w}_2 + \cdots + a_{km}\vec{w}_k
\end{aligned}$$

This is true by definition of span (all in $\vec{v}$ are in the span of W).

Pick a vector $\vec{w} \in W$.

$$\vec{w} = r_1\vec{v}_1 + r_2\vec{v}_2 + \cdots + r_m\vec{v}_m \in \text{sp}\{\vec{v}_1 \cdots \vec{v}_m\}$$

By the definition of span. Now, we can rewrite w by replacing $\vec{v}_1$ by the first equation.

$$\begin{aligned}
\vec{w} &= r_1(a_{11}\vec{w}_1 + a_{21}\vec{w}_2 + \cdots a_{k1}\vec{w}_k) + r_2(a_{12}\vec{w}_1 + a_{22}\vec{w}_2 + \cdots + a_{k2}\vec{w}_k) + \cdots + r_n(a_{1m}\vec{w}_1 + a_{2m}\vec{w}_2 \\
&\quad + \cdots + a_{km}\vec{w}_k) \\
&= (r_1a_{11} + r_2a_{12} + \cdots + r_ma_{1m})\vec{w}_1 + \cdots + (r_1a_{k1} + r_2a_{k2} + \cdots + r_ma_{km})\vec{w}_k
\end{aligned}$$

If $\vec{w} = 0$, then $A\vec{r} = \vec{0}$, where A is a matrix formed by the column vectors of $\vec{v}_1 \cdots \vec{v}_m$. We know that $\vec{v}_1 \cdots \vec{v}_m$ is linearly independent, meaning the matrix has a unique and trivial solution. This means $\vec{r} = 0$. This means

$$\begin{aligned}
&r_1a_{11} + r_2a_{12} + \cdots + r_{m1}a_{1m} = 0 \\
&\vdots \\
&r_1a_{k1} + r_2a_{k2} + \cdots + r_ma_{km} = 0
\end{aligned}$$

This system only contains one solution, and that solution must be the trivial solution of $\vec{r} = 0$. However, you have k equations and m unknowns. Recall that we assumed $k < m$, meaning the number of equations is less than the number of unknowns, and you cannot have a unique solution that way – you would have infinitely many solutions.

~~You are supposed to have infinitely many $\vec{r}$ to satisfy $r_1\vec{v}_1 + r_2\vec{v}_2 + \cdots + r_m\vec{v}_m = \vec{0}$. Therefore $\vec{v}_1 \cdots \vec{v}_m$ are linearly independent.~~